US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12406713
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Debashis; Punyashloka et al.

Probabilistic computing devices based on stochastic switching in a ferroelectric field-effect transistor

Abstract

A pbit device, in one embodiment, includes a first field-effect transistor (FET) that includes a source region, a drain region, a source electrode on the source region, a drain electrode on the drain region, a channel region between the source and drain regions, a dielectric layer on a surface over the channel region, an electrode layer above the dielectric layer, and a ferroelectric (FE) material layer between the dielectric layer and the electrode layer. The pbit device also includes a second FET comprising a source electrode, a drain electrode, and a gate electrode. The drain electrode of the second FET is connected to the drain electrode of the first FET.

Inventors: **Debashis; Punyashloka (Hillsboro, OR), Nikonov; Dmitri Evgenievich (Beaverton, OR), Li; Hai (Portland, OR), Lin; Chia-Ching (Portland, OR), Kim; Raseong (Portland, OR), Gosavi; Tanay A. (Portland, OR), Penumatcha; Ashish Verma (Beaverton, OR), Avci; Uygar E. (Portland, OR), Radosavljevic; Marko (Portland, OR), Young; Ian Alexander (Olympia, WA)**

Applicant: **Intel Corporation** (Santa Clara, CA)

Family ID: **85228493**

Assignee: **Intel Corporation (Santa Clara, CA)**

Appl. No.: **17/409483**

Filed: **August 23, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20230058938 A1	Feb. 23, 2023

Publication Classification

Int. Cl.: **G11C11/22** (20060101); **H10B51/30** (20230101)

U.S. Cl.:

CPC **G11C11/2275** (20130101); **G11C11/223** (20130101); **G11C11/2273** (20130101); **H10B51/30** (20230201); G11C11/2293 (20130101)

Field of Classification Search

CPC: G11C (11/2275); G11C (11/223); G11C (11/2273); G11C (11/2293); H10B (51/30)

USPC: 365/145

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9935105	12/2017	Yabuuchi	N/A	H10D 89/10
11342927	12/2021	Lim	N/A	H03L 7/0995
2008/0024181	12/2007	Wada	327/161	G11C 29/023
2019/0273087	12/2018	Morris	N/A	H10D 30/6735
2024/0243714	12/2023	Pattipaka	N/A	H04B 1/04

OTHER PUBLICATIONS

Deng, Shan, et al. “A Comprehensive Model for Ferroelectric FET Capturing the Key Behaviors: Scalability, Variation, stochasticity, and Accumulation.” 2020 IEEE Symposium on VLSI Technology. IEEE, 2020. cited by applicant

Mulaosmanovic, Halid, Thomas Mikolajick, and Stefan Slesazeck. “Random number generation based on ferroelectric switching.” IEEE Electron Device Letters 39.1 (2017): 135-138. cited by applicant

Shin, Young-Han, et al. “Nucleation and growth mechanism of ferroelectric domain-wall motion.” Nature 449.7164 (2007): 881-884. cited by applicant

Primary Examiner: Yoha; Connie C

Attorney, Agent or Firm: Alliance IP, LLC

Background/Summary

BACKGROUND

(1) Probabilistic computing has been shown to provide large power, performance, and area improvements compared to traditional complementary metal-oxide-semiconductor (CMOS) devices for artificial intelligence (AI)-based compute applications, such as optimization and sampling. The key enabler for probabilistic computing is a hardware called a “pbit” that utilizes intrinsic physics to generate binary random numbers with tunable probability. This intrinsic property of the pbit enables parallel and autonomous minimization of a cost function that can be

designed to represent the solution to computationally hard problems of practical interest, such as the traveling salesman problem, integer factorization, Bayesian networks, etc.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1**A-**1**C illustrate an example probabilistic pbit device and operational characteristics of the pbit device.

(2) FIGS. **2**A-**2**B illustrate an example Ising network and its associated characteristics.

(3) FIG. **3** illustrates an example pbit device comprising an n-channel FeFET in accordance with embodiments of the present disclosure.

(4) FIG. **4**A illustrates an example pulse cycle that may be applied as input to a FeFET-based pbit device in accordance with embodiments of the present disclosure.

(5) FIG. **4**B illustrates an example FE switching probability for a FeFET-based pbit device as a function of input pulse width or height.

(6) FIG. **5** illustrates another example FeFET-based pbit device that includes the device of FIG. **3**.

(7) FIG. **6** illustrates an example timing diagram for the FeFET-based pbit device of FIG. **5** in accordance with embodiments of the present disclosure.

(8) FIG. **7** illustrates an example pbit device comprising a p-channel FeFET in accordance with embodiments of the present disclosure.

(9) FIG. **8** illustrates another example pbit device comprising an n-channel FeFET in accordance with embodiments of the present disclosure.

(10) FIG. **9** illustrates an example resistive circuit for connecting neighboring pbit devices of a computational network in accordance with embodiments of the present disclosure.

(11) FIG. **10** illustrates an example capacitive circuit for connecting neighboring pbit devices of a computational network in accordance with embodiments of the present disclosure.

(12) FIG. **11** is a flow diagram illustrating an example process **1100** of operating a FeFET of a pbit device in accordance with embodiments of the present disclosure.

(13) FIG. **12** is a top view of a wafer and dies that may be included in a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

(14) FIG. **13** is a cross-sectional side view of an integrated circuit device that may be included in a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

(15) FIGS. **14**A-**14**D are perspective views of example planar, FinFET, gate-all-around, and stacked gate-all-around transistors.

(16) FIG. **15** is a cross-sectional side view of an integrated circuit device assembly that may include a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

(17) FIG. **16** is a block diagram of an example electrical device that may include a microelectronic assembly, in accordance with any of the embodiments disclosed herein.

DETAILED DESCRIPTION

(18) Probabilistic computing has been shown to provide large power, performance, and area improvements compared to traditional complementary metal-oxide-semiconductor (CMOS) devices for artificial intelligence (AI)-based compute applications, such as optimization and sampling. The key enabler for probabilistic computing is a hardware called a “pbit” that utilizes intrinsic physics to generate binary random numbers with tunable probability. This intrinsic property of the pbit enables parallel and autonomous minimization of a cost function that can be designed to represent the solution to computationally hard problems of practical interest, such as the traveling salesman problem, integer factorization, Bayesian networks, etc.

(19) Previous solutions have discussed magnetic tunnel junctions (MTJs) with a thermally unstable free layer for providing the intrinsic stochasticity, owing to its technology maturity, room

temperature operation and compatibility with CMOS BEOL. In such devices, the unstable MTJ in series with a NMOS transistor forms the tunable random number generator. A CMOS inverter is utilized to amplify the stochastic voltage fluctuations at the NMOS drain node. This forms the complete spintronic pbit. However, MTJs have a relatively small output ratio of around 100% that comes from the change in the device resistance as the magnetization of the free layer switches between parallel and antiparallel orientation with respect to the reference layer magnetization. Therefore, the CMOS inverter is required to amplify the signal, but can introduce its own source of noise and additional device footprint. Moreover, being magnetic devices, the spintronic pbits are prone to external magnetic attacks and present a challenge to minimize the stray magnetic fields coming from neighboring devices within such a chip.

(20) In contrast, aspects of the present disclosure utilize new pbit devices that incorporate FE-based field-effect transistor (FEFET or FeFET) designs. In such devices, a FE material may be used between a gate electrode and the channel of the FET. The intrinsically probabilistic/stochastic nature of the FE polarization switching (as a function of voltage pulse amplitude or duration) can be used in certain embodiments to encode the probability of the pbit device. When a series of voltage pulses are applied to the gate of the FeFET, its threshold voltage can switch randomly between a large ($V_{T,high}$) and small ($V_{T,low}$) value, with a probability that depends on the gate voltage pulse height, width, or the voltage of another terminal.

(21) FE-based pbit devices may have one or more advantages over its traditional spintronics counterpart. For example, FE-based pbit devices may have an output ratio that is orders of magnitude larger than magnetic pbit devices, as it is derived from the change in the transistors drain current as opposed to the tunnel magnetoresistance effect. In addition, FE-based pbit devices may be robust against magnetic attacks or internal magnetic fields produced within the chip, since the core element is the ferroelectric polarization switching, which is a charge-based phenomenon. As another example, FE switching is potentially faster than FM switching. Hence, the speed of operation of the FE-based pbit device may be faster when compared to its spintronics counterpart. As yet another example, since the pbit device output only changes after the application of a gate voltage pulse, its operation speed is more robust against device-to-device variation compared to spontaneously fluctuating pbits whose magnetic parameters determine the device operation speed.

(22) FIGS. **1**A-**1**C illustrate an example probabilistic pbit device **100** and operational characteristics of the pbit device. In particular, FIG. **1**A illustrates a probabilistic pbit device that accepts an input I_i and produces a probabilistic output m_i. The probabilistic characteristics of the pbit device **100** are shown in FIGS. **1**B-**1**C. For instance, FIG. **1**B illustrates instantaneous random fluctuations of the output m_i for three different inputs I_i (−1, 0, 1). As shown, the output signal m_i fluctuates in a seemingly random nature for the same input signal I_i; however, the long term average of the output signal m_i demonstrates a probabilistic distribution as shown in FIG. **1**C. FIG. **1**B illustrates a histogram of example output states, showing that the input changes the probability distribution of the probabilistic output. As shown in FIG. **1**C, the long-term average of the output m_i as a function of I_i shows a sigmoidal response.

(23) This probabilistic nature of the pbit device **100** can be utilized to design devices that model certain problems of interest, e.g., optimization and/or sampling problems. For instances, a set of pbit devices can be interconnected in a networked way to model such problems. As an example, a set of pbit devices with responses modeled by the neuron equation:

$$m_i=\mathrm{sign}[\tanh I_i-rnd_{(-1,1)}]$$

can be interconnected through weights W_{ij} as shown in the synapse equation:

$$I_i=\beta\Sigma W_{ij}\times m_j$$

By designing an appropriate weight matrix [W_{ij}], many problems of practical interest such as optimization and sampling can be mapped onto this pbit-based hardware device.

(24) FIGS. **2**A-**2**B illustrate an example Ising network **200** and its associated characteristics. The

example Ising network **200** shown in FIG. **2**A may be used, in certain embodiments, to solve combinatorial optimization problems or other types of problems. In the example shown, the Ising network **200** includes a set of interconnected pbit devices **202** with particular spins (S), which can take values of +1 (up) or −1 (down) randomly. Each spin is connected to its neighbors through weighted interconnections that determines the probability of the spin being in the +1 or −1 state. Overall, the network has an energy E, which depends on the configuration of the spins. For instance, the pbit device **202**A has a spin S_A that is −1/down, pbit device **202**B has a spin S_B that is +1/up, and the pbit device **202**C has a spin S_C that is −1/down. The pbit devices **202** are connected through interconnection strengths or weights **204** (J). In the example shown, the interconnection strength **204**A between the pbit devices **202**A, **202**B is shown as J_{AB}, and the interconnection strength **204**B between the pbit devices **202**B, **202**C is shown as J_{BC}. Although not labeled in FIG. **2**A, the other pbit devices have similar interconnection strengths, each of which may be different from one another.

(25) FIG. **2**B illustrates a plot **250** of the network energy of the Ising network **200** as a function of the spin state of the network. Overall, the network has an energy E, which depends on the configuration of the spins. The energy E is represented by the equation shown in FIG. **2**B, where Jij is the interconnection strength between the i-th and the j-th spin (Si and Sj), and hi is the bias strength. Each spin of the Ising network **200** can be represented in hardware with a ferroelectric-based pbit device, such as those shown and described herein. The example Ising network **200** follows the Boltzmann distribution, with the highest probability of being in the lowest energy state that encodes the solution of the optimization problem. The output of the Ising network **200** may be read and used as appropriate (e.g., analyzed, used as an input to another problem, etc.).

(26) FIG. **3** illustrates an example pbit device **300** comprising an n-channel FeFET in accordance with embodiments of the present disclosure. The example pbit device **300** may be used in a hardware computational network, such as an Ising network (e.g., **200** of FIG. **2**A), that is used to model a probabilistic algorithm. The example pbit device **300** includes a n-channel FeFET **310** in series with a PMOS transistor **320**. The source of the PMOS transistor **320** is connected to a supply voltage V_{DD} and the drain of the PMOS transistor is connected to the drain of the n-channel FeFET **310**. The source of the n-channel FeFET **310** is connected to ground. The gate of the PMOS transistor **320** is connected to the opposite of a read (RD) clock signal (i.e., RD_bar). The RD clock signal may be a logic 1 signal when a read pulse is active, as described below.

(27) The example n-channel FeFET **310** includes n-doped source/drain regions **315**/**316** in a substrate **317** (which may be p-doped in some instances), with electrodes formed on each source/drain region. The FeFET **310** further includes a dielectric layer **314** formed over a channel region **318** of the FeFET **310**, a FE material **313** formed on the dielectric layer **314** and a gate electrode layer **312** formed on the FE material **313**. The dielectric layer **314** may include any suitable dielectric material as described below, which in some instances, may include silicon and oxygen (e.g., silicon oxide (e.g., SiO_2)). The FE material **313** may include, in some instances, hafnium and oxygen (e.g., hafnium oxide (e.g., HfO_2)). The gate electrode layer **312** may include one or more metal-based layers, which may include one or more of polycrystalline silicon (Poly-Si), titanium, or nitrogen (e.g., a TiN layer and/or a Poly-Si layer). Although shown in FIG. **3** as a planar FET, certain embodiments may utilize non-planar FETs.

(28) To operate the example pbit device **300**, a series of n pulse cycles may be applied as input to node **302** (V_G) coupled to the gate electrode **312**. Each pulse cycle may be configured as shown in FIG. **4**A and described further below, including a reset pulse, write pulse, and read pulse. The switching probability P_{SW} of the ferroelectric polarization of the FE material **313** in the FeFET **310** may be a function of write pulse height or width, as shown in FIG. **4**B. Based on the switching probability P_{SW}, the output V_X may be low or “0” state with a probability of ($n \times P_{SW}$) and high or “1” state with a probability of ($n \times (1-P_{SW})$) as shown.

(29) The FE material **313** may be polarized based on the following phenomenon. When an external

electric field is applied across a ferroelectric (FE) film in a direction opposite its polarization, some FE domains reverse their polarization to form a critical nucleus, which then leads to the polarization switching of the entire FE film due to domain growth. The initial domain nucleation step behaves as a Poisson process and leads to the inherent stochasticity of the FE polarization switching. As the area of the FE film is scaled, this switching becomes discrete, eventually becoming a binary event where the FE polarization points either "up" or "down" with a probability that is dependent on the applied field. Thus, the polarization of the FE material **313** within the FeFET may be used to indicate a binary state/spin of the pbit device **300**.
(30) FIG. **4**A illustrates an example pulse cycle **400** that may be applied as input to a FeFET-based pbit device (e.g., the pbit device **300** of FIG. **3**) in accordance with embodiments of the present disclosure. The example pulse cycle **400** includes a reset pulse **410** (negative in this example, which is for the n-channel FeFET **310** of FIG. **3**), a write pulse **420** (positive in this example for the n-channel FeFET **310** of FIG. **3**), and a read pulse **430** (also positive in this example for the n-channel FeFET **310** of FIG. **3**). The reset pulse **410** may be relatively large so that it deterministically sets the polarization of the FE material of the FeFET (e.g., **313**). For instance, referring to the example FeFET **310** of FIG. **3**, after the reset pulse **410**, the FE polarization will be forced to the positive direction (pointing up in FIG. **3**).
(31) During the write pulse **420**, the FE polarization may switch from positive to negative (pointing down in the example shown in FIG. **3**) with a probability that is determined by the height or the width of the write pulse **420**. The direction of the FE polarization sets the V.sub.T of the FeFET. In certain embodiments, e.g., as described further below, the magnitude or the width of the write pulse **420** may be based on outputs of other FeFET-based pbit devices.
(32) During the read pulse **430**, the RD_bar signal will be input to the PMOS transistor **320**. Thus, depending upon the V.sub.T of FeFET (which is determined probabilistically from the write pulse **420**), the read pulse **430** will cause the voltage at the output node (e.g., Vx at output node **330** in FIG. **3**) will be either pulled to V.sub.DD (which may be considered as the −1/down state of the pbit) or ground (which may be considered as the +1/up state of the pbit). Thus, in certain embodiments the magnitude of the read pulse **430** may be set to be a voltage that is between the high V.sub.T and low V.sub.T of the FeFET.
(33) FIG. **5** illustrates another example FeFET-based pbit device **500** that includes the device **300** of FIG. **3**. In the example shown, the input node **302** of the pbit device **300** is coupled to a pulse generation circuit **510**, and the output node **330** of the pbit device **300** is coupled to a D flip-flop circuit **520**. The example pulse generation circuit **510** includes a first NMOS transistor **511** whose gate is connected to a read (RD) clock signal, drain is connected to a V.sub.read voltage source, and source is connected to the input node **302**, a second NMOS transistor **512** whose gate is connected to a reset (RST) clock signal, drain is connected to a V.sub.reset voltage source, and source is connected to the input node **302**, and a third NMOS transistor **513** whose gate is connected to a write (WR) clock signal, drain is connected to an output node **514** for of one or more neighboring pbit devices (e.g., in an Ising network), and source is connected to the input node **302**. The RD, RST, and WR clock signals may be implemented at different times from one another, e.g., in the manner shown in FIG. **6** and described further below. Further, the example D flip-flop circuit **520** may take the voltage V.sub.X at node **330** and generate a logic 1/0 output voltage V.sub.out (and it's opposite V.sub.out_bar), e.g., as shown in FIG. **6** and described further below. Example connections to multiple neighboring pbit devices for node **514** are shown in FIGS. **9-10** and described further below. It will be understood that the pulse generation circuit **510** may be used to provide the respective pulses to multiple pbit devices (rather than just the one as shown in FIG. **5**), e.g., in computational networks (e.g., Ising networks) that include multiple pbit devices as described herein.
(34) FIG. **6** illustrates an example timing diagram for the FeFET-based pbit device **500** of FIG. **5** in accordance with embodiments of the present disclosure. The example timing diagram includes a

chart **610** showing the RD, RST, and WR clock signals, a chart **620** showing the voltage at the input node **302** (V.sub.G) of the pbit device **300**, a chart **630** showing the voltage at the output node **330** (V.sub.X) of the pbit device **300**, and a chart **640** showing the logic signal voltage (V.sub.out) produced by the D flip-flop circuit **520** based on the voltage at the output node **330**.
(35) In the example shown in FIG. **6**, there are 5 full cycles (e.g., the n cycles described above) and ⅔ of a 6.sup.th cycle illustrated. In each cycle, the RST clock signal (a logic 1 signal) is activated for a t.sub.reset period, followed by the WR clock signal (a logic 1 signal) for a t.sub.write period, and then followed by the RD clock signal (a logic 1 signal) for a t.sub.read period. During the t.sub.reset period of each cycle, a large negative voltage is applied at the node **302** to deterministically set the polarization of the FE material of the FeFET **310** (shown in FIG. **6** by the dotted line, P).
(36) During the t.sub.write period, a voltage V.sub.write is applied at the node **302**. The V.sub.write voltage may be based on one or more outputs of neighboring pbit devices in a probabilistic modeling computational network, such as an Ising network. In some instances, the voltage may be a weighted sum of neighboring output voltage, e.g., generated through one of the summing circuits described further below with respect to FIGS. **9-10**. Due to the probabilistic nature of the pbit network, the V.sub.write voltage may vary each cycle, as shown in FIG. **6**. Depending on the V.sub.write voltage applied during the t.sub.write period, the polarization of the FE material within the FeFET may flip or not from the polarization set by the reset pulse. For instance, in the first cycle shown in FIG. **6**, the voltage applied during the t.sub.write period is large enough to flip the polarization; in contrast, the voltage applied during the t.sub.write period of the second cycle shown in FIG. **6** is not large enough to flip the polarization.
(37) Next, during the t.sub.read period, a particular voltage V.sub.read is applied to the node **302** to determine the polarization of the FeFET **310**. The voltage V.sub.read may be between the high and low threshold voltages that are inherent to the FeFET, and are based on the state of the polarization of the FE material in the FeFET. The voltage V.sub.read may cause the voltage at the node **330** (V.sub.X) to stay low, e.g., when the polarization is flipped, or to go high, e.g., when the polarization is not flipped. Based on this, the voltage V.sub.out may change. However, the reading of V.sub.out during the tread may be delayed by one cycle due to the D flip-flop circuit **520**. That is, the reading of the V.sub.out in an n-th cycle may represent the state written to the pbit device in the (n−1)-th cycle. In the example shown, a positive polarization (caused by a higher magnitude positive V.sub.G) is considered as a logic 0 signal while a negative polarization (caused by a lower magnitude positive V.sub.G) is considered as a logic 1 signal.
(38) For example, in the first cycle shown in FIG. **6**, the polarization is flipped during the t.sub.write period, which causes the voltage V.sub.x to stay relatively low in the first cycle. This low V.sub.x state, however, is captured in the second cycle shown during the t.sub.read period as shown in FIG. **6**. Likewise, in the second cycle shown in FIG. **6**, the polarization is not flipped during the t.sub.write period, which causes the voltage V.sub.x to go relatively high in the second cycle. This higher V.sub.x state is captured in the third cycle shown during the t.sub.read period as shown in FIG. **6**.
(39) FIG. **7** illustrates an example pbit device **700** comprising a p-channel FeFET in accordance with embodiments of the present disclosure. The example pbit device **700** may be used in a hardware computational network, such as an Ising network (e.g., **200** of FIG. **2**A), that is used to model a probabilistic algorithm. The example pbit device **700** includes a p-channel FeFET **710** in series with a NMOS transistor **720**. The source of the NMOS transistor **720** is connected to ground while the drain of the NMOS transistor is connected to the drain of the p-channel FeFET **710**. The source of the p-channel FeFET **310** is connected to the supply voltage V.sub.DD. The gate of the NMOS transistor **720** is connected to the read (RD) clock signal, which is described above.
(40) The example p-channel FeFET **710** includes p-doped source/drain regions **715**/**716** in a substrate **717** (which may be n-doped in some instances), with electrodes formed on each

source/drain region. The FeFET **710** further includes a dielectric layer **714** formed over a channel region **718** of the FeFET **710**, a FE material **713** formed on the dielectric layer **714** and a gate electrode layer **712** formed on the FE material **713**. The dielectric layer **714** may include any suitable dielectric material as described below, which in some instances, may include silicon and oxygen (e.g., silicon oxide (e.g., SiO.sub.2)). The FE material **713** may include, in some instances, hafnium and oxygen (e.g., hafnium oxide (e.g., HfO.sub.2)). The gate electrode layer **712** may include one or more metal-based layers, which may include one or more of polycrystalline silicon (Poly-Si), titanium, or nitrogen (e.g., a TiN layer and/or a Poly-Si layer). Although shown in FIG. **7** as a planar FET, certain embodiments may utilize non-planar FETs.
(41) The example pbit device **700** may be operated in similar manner to the pbit device **300** of FIG. **3**, but with input/output voltages of opposite polarity. For instance, the input n cycles may be similar to those shown in FIG. **4**A, but with opposite polarity. That is, the reset pulse may be a highly positive voltage pulse to deterministically set the polarization of the FE material **713**, and the write pulse and read may be negative voltages of smaller magnitude than the reset pulse. The switching probability P.sub.SW of the ferroelectric polarization of the FE material **713** in the FeFET **710** may also be a function of write pulse height or width. Based on the switching probability P.sub.SW, the output V.sub.X may be low/“0” state with a probability of (n×(1−P.sub.SW)) and high/“1” with a probability of (n×P.sub.SW) as shown.
(42) FIG. **8** illustrates another example pbit device **800** comprising an n-channel FeFET **810** in accordance with embodiments of the present disclosure. The example pbit device **800** may be used in a hardware network, such as an Ising network (e.g., **200** of FIG. **2**A), that is used to model a probabilistic algorithm. The example pbit device **800** includes a n-channel FeFET **810** in series with a PMOS transistor **820**. The source of the PMOS transistor **820** is connected to a supply voltage V.sub.DD and the drain of the PMOS transistor is connected to the drain of the n-channel FeFET **810**. The source of the n-channel FeFET **810** is connected to ground. The gate of the PMOS transistor **820** is connected to the opposite of a read (RD) clock signal (i.e., RD_bar). The RD clock signal may be a logic 1 signal when a read pulse is active, as described above.
(43) The example n-channel FeFET **810** includes n-doped source/drain regions **815**/**816** in a substrate **817** (which may be p-doped in some instances), with electrodes formed on each source/drain region. The FeFET **810** further includes a dielectric layer **814** formed over a channel region **818** of the FeFET **810**, a first electrode layer **819** formed on the dielectric layer **814**, a FE material **813** formed on the first electrode layer **819**, and a second electrode layer **812** formed on the FE material **813**. The dielectric layer **814** may include any suitable dielectric material as described below, which in some instances, may include silicon and oxygen (e.g., silicon oxide (e.g., SiO.sub.2)). The FE material **813** may include, in some instances, hafnium and oxygen (e.g., hafnium oxide (e.g., HfO.sub.2)). The first or second electrode layer **812** may include one or more metal-based layers, which may include one or more of polycrystalline silicon (Poly-Si), titanium, or nitrogen (e.g., a TiN layer and/or a Poly-Si layer). Although shown in FIG. **8** as a planar FET, certain embodiments may utilize non-planar FETs.
(44) The input node **802** of the pbit device **800** is coupled to a pulse generation circuit **840**, and the output node **830** of the pbit device **800** is coupled to a D flip-flop circuit **850**. The example pulse generation circuit **840** includes a first NMOS transistor **841** whose gate is connected to a read (RD) clock signal, drain is connected to a V.sub.read voltage source, and source is connected to the input node **802**, a second NMOS transistor **842** whose gate is connected to a reset (RST) clock signal, drain is connected to a V.sub.reset voltage source, and source is connected to the input node **802**, and a third NMOS transistor **843** whose gate is connected to a write (WR) clock signal, drain is connected to a V.sub.bias voltage source, and source is connected to the input node **802**. The first electrode layer **819** is connected to an output node **845** for of one or more neighboring pbit devices (e.g., in an Ising network) through a fourth NMOS transistor **844** whose gate is connected to the WR clock signal. By only applying a voltage to the first electrode layer **819** during the write phase

(when the WR signal is active), the first electrode layer **819** may be floated during the read and reset phases. The RD, RST, and WR clock signals may be implemented at different times from one another, e.g., in the manner shown in FIG. **6** and described above. As above, the example D flip-flop circuit **850** may take the voltage V.sub.X at node **830** and generate a logic 1/0 output voltage V.sub.out (and it's opposite V.sub.out_bar), e.g., as shown in FIG. **6** and described above. Example connections to multiple neighboring pbit devices for node **845** are shown in FIGS. **9-10** and described further below. It will be understood that the pulse generation circuit **840** may be used to provide the respective pulses to multiple pbit devices (rather than just the one as shown in FIG. **8**), e.g., in computational networks (e.g., Ising networks) that include multiple pbit devices as described herein.

(45) The example pbit device **800** works in a similar manner to the pbit devices discussed above. However, in the example shown, the probability of the FE polarization switching in the FE material **813** is controlled through the voltage applied to the first electrode layer **819**. Therefore, in the input voltage sequence, a write pulse height and width are kept constant based on the V.sub.bias voltage applied at **802**, and the voltage at **845** (which is based on the voltage of neighboring nodes in a network such as an Ising network) determines the overall voltage drop across the FE material layer **813** and accordingly controls the probability of the FE switching. Therefore, the pulse generation circuitry **840** of the pbit device **800** may be made slightly simpler compared to the pulse generation circuitry **510** of the pbit device **500**. In addition, one advantage of the pbit device **800** is that having a symmetric metal/FE/metal stack in **812/813/819** may improve the endurance of the gate stack.

(46) FIG. **9** illustrates an example resistive circuit **900** for connecting neighboring pbit devices of a computational network in accordance with embodiments of the present disclosure. In the example shown, outputs **910** from a set of pbit devices (e.g., from the D flip-flop circuit **520** of FIG. **5** or the D flip-flop circuit **850** of FIG. **8**) are weighted and summed using a resistive array **920** (e.g., memristive crossbar or multi-level MRAM array with reconfigurable resistance values), and the summing amplifier circuit **930**, and provided as the input to the next neighboring pbit device (e.g., at node **514** of the pbit device **500** of FIG. **5** or node **845** of the pbit device **800** of FIG. **8**).

(47) The values of the resistors in the array **920** may be adjusted to apply different weights to the outputs of the neighboring pbit devices. In the example shown, there are different resistors connected to the respective outputs of the D flip-flop circuits of the neighboring pbit devices. The values of these (e.g., w.sub.i,p and w.sub.i,n) may be different from one another. For instance, one of the resistor values may be infinite (or effectively infinite) where only one of the up/down output voltage signals is to be summed by the circuit **930**. The w.sub.i,p value may apply a weight to the up output, while the w.sub.i,n value may apply a weight to the down output. The summing amplifier circuit **930** includes an opamp in a negative feedback configuration, with a reference voltage applied at the positive input of the opamp. The reference voltage V.sub.ref may be a voltage that is between the supply voltage V.sub.DD and ground (e.g., V.sub.DD/2). The summing amplifier circuit **930** may effectively sum the currents provided at the input node and provide a corresponding output voltage.

(48) FIG. **10** illustrates an example capacitive circuit **1000** for connecting neighboring pbit devices of a computational network in accordance with embodiments of the present disclosure. In the example shown, outputs **1010** from a set of pbit devices (e.g., from the D flip-flop circuit **520** of FIG. **5** or the D flip-flop circuit **850** of FIG. **8**) are weighted and summed using a binary weighted array of switch capacitors **1020** and a summing amplifier circuit **1030**, and provided as the input to the next neighboring pbit device (e.g., at node **514** of the pbit device **500** of FIG. **5** or node **845** of the pbit device **800** of FIG. **8**).

(49) The values of the capacitors in the array **1020** may be adjusted to apply different weights to the outputs of the neighboring pbit devices. In the example shown, there are different capacitors connected to the respective outputs of the D flip-flop circuits of the neighboring pbit devices. The values of these (e.g., w.sub.i,p and w.sub.i,n) may be different from one another. For instance, one

of the capacitor values may be zero where only one of the up/down output voltage signals is to be summed by the circuit **1030**. The w.sub.i,p value may apply a weight to the up output, while the w.sub.i,n value may apply a weight to the down output. The summing amplifier circuit **1030** includes an opamp in a negative feedback configuration (with a switched capacitor coupled to the WR_bar clock signal), with a reference voltage applied at the positive input of the opamp. The reference voltage V.sub.ref may be a voltage that is between the supply voltage V.sub.DD and ground (e.g., V.sub.DD/2). The summing amplifier circuit **930** may effectively sum the voltages provided at the input node and provide a corresponding output voltage.

(50) FIG. **11** is a flow diagram illustrating an example process **1100** of operating a FeFET of a pbit device in accordance with embodiments of the present disclosure. The example process may include additional or different operations, and the operations may be performed in the order shown or in another order. In some cases, one or more of the operations shown in FIG. **11** are implemented as processes that include multiple operations, sub-processes, or other types of routines. In some cases, operations can be combined, performed in another order, performed in parallel, iterated, or otherwise repeated or performed another manner.

(51) At **1102**, a reset pulse is applied to the FeFET of the pbit device. The reset pulse may be a voltage with a magnitude that deterministically set a polarization state of the FE material layer of the FeFET. At **1104**, a write pulse is applied to the FeFET of the pbit device. The write pulse may be a voltage with a magnitude that is based on an output voltage of at least one other FeFET, e.g., FeFETs of other interconnected pbit devices. The output voltages of the other FeFETs may be based on a polarization state of the respective FE material layers of the FeFETs, as described above. At **1106**, a read pulse is applied to the FeFET of the pbit device. The read pulse may be a voltage with a magnitude that is between a first threshold voltage of the FET based on a first polarization state of the FE material layer (e.g., positive polarization) and a second threshold voltage of the FET based on a second polarization state of the FE material layer (e.g., negative polarization). At **1108**, a polarization state of the pbit device is determined. The polarization state may be determined based on a voltage detected at an output node of the pbit device, e.g., at the voltage at the drain electrode of the FeFET **310** of FIG. **3** (V.sub.X).

(52) At **1110**, it is determined whether there have been a particular number (N) of pulse cycles applied to the FeFET of the pbit device. If not, the operations **1102**, **1104**, **1106**, and **1008** are performed again. If N pulse cycles have been performed, then at **1112**, a statistical probability of the polarization state of the pbit device is determined. The statistical probability may be used to determine a solution of a computational problem, e.g., as described above.

(53) FIG. **12** is a top view of a wafer **1200** and dies **1202** that may include any of the FeFET-based pbit devices disclosed herein. The wafer **1200** may be composed of semiconductor material and may include one or more dies **1202** having integrated circuit structures formed on a surface of the wafer **1200**. The individual dies **1202** may be a repeating unit of an integrated circuit product that includes any suitable integrated circuit. After the fabrication of the semiconductor product is complete, the wafer **1200** may undergo a singulation process in which the dies **1202** are separated from one another to provide discrete “chips” of the integrated circuit product. The die **1202** may include one or more transistors (e.g., some of the transistors **1340** of FIG. **13**, discussed below), supporting circuitry to route electrical signals to the transistors, passive components (e.g., signal traces, resistors, capacitors, or inductors), and/or any other integrated circuit components. In some embodiments, the wafer **1200** or the die **1202** may include a memory device (e.g., a random access memory (RAM) device, such as a static RAM (SRAM) device, a magnetic RAM (MRAM) device, a resistive RAM (RRAM) device, a conductive-bridging RAM (CBRAM) device, etc.), a logic device (e.g., an AND, OR, NAND, or NOR gate), or any other suitable circuit element. Multiple ones of these devices may be combined on a single die **1202**. For example, a memory array formed by multiple memory devices may be formed on a same die **1202** as a processor unit (e.g., the processor unit **1602** of FIG. **16**) or other logic that is configured to store information in the memory

devices or execute instructions stored in the memory array. Various ones of the FeFET-based pbit devices disclosed herein may be manufactured using a die-to-wafer assembly technique in which some dies are attached to a wafer **1200** that include others of the dies, and the wafer **1200** is subsequently singulated.

(54) FIG. **13** is a cross-sectional side view of an integrated circuit device **1300** that may include any of the FeFET-based pbit devices disclosed herein. One or more of the integrated circuit devices **1300** may be included in one or more dies **1202** (FIG. **12**). The integrated circuit device **1300** may be formed on a die substrate **1302** (e.g., the wafer **1200** of FIG. **12**) and may be included in a die (e.g., the die **1202** of FIG. **12**). The die substrate **1302** may be a semiconductor substrate composed of semiconductor material systems including, for example, n-type or p-type materials systems (or a combination of both). The die substrate **1302** may include, for example, a crystalline substrate formed using a bulk silicon or a silicon-on-insulator (SOI) substructure. In some embodiments, the die substrate **1302** may be formed using alternative materials, which may or may not be combined with silicon, that include, but are not limited to, germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, or gallium antimonide. Further materials classified as group II-VI, III-V, or IV may also be used to form the die substrate **1302**. Although a few examples of materials from which the die substrate **1302** may be formed are described here, any material that may serve as a foundation for an integrated circuit device **1300** may be used. The die substrate **1302** may be part of a singulated die (e.g., the dies **1202** of FIG. **12**) or a wafer (e.g., the wafer **1200** of FIG. **12**).

(55) The integrated circuit device **1300** may include one or more device layers **1304** disposed on the die substrate **1302**. The device layer **1304** may include features of one or more transistors **1340** (e.g., metal oxide semiconductor field-effect transistors (MOSFETs)) formed on the die substrate **1302**. The transistors **1340** may include, for example, one or more source and/or drain (S/D) regions **1320**, a gate **1322** to control current flow between the S/D regions **1320**, and one or more S/D contacts **1324** to route electrical signals to/from the S/D regions **1320**. The transistors **1340** may include additional features not depicted for the sake of clarity, such as device isolation regions, gate contacts, and the like. The transistors **1340** are not limited to the type and configuration depicted in FIG. **13** and may include a wide variety of other types and configurations such as, for example, planar transistors, non-planar transistors, or a combination of both. Non-planar transistors may include FinFET transistors, such as double-gate transistors or tri-gate transistors, and wrap-around or all-around gate transistors, such as nanoribbon, nanosheet, or nanowire transistors.

(56) FIGS. **14**A-**14**D are simplified perspective views of example planar, FinFET, gate-all-around, and stacked gate-all-around transistors. The transistors illustrated in FIGS. **14**A-**14**D are formed on a substrate **1416** having a surface **1408**. Isolation regions **1414** separate the source and drain regions of the transistors from other transistors and from a bulk region **1418** of the substrate **1416**.

(57) FIG. **14**A is a perspective view of an example planar transistor **1400** comprising a gate **1402** that controls current flow between a source region **1404** and a drain region **1406**. The transistor **1400** is planar in that the source region **1404** and the drain region **1406** are planar with respect to the substrate surface **1408**.

(58) FIG. **14**B is a perspective view of an example FinFET transistor **1420** comprising a gate **1422** that controls current flow between a source region **1424** and a drain region **1426**. The transistor **1420** is non-planar in that the source region **1424** and the drain region **1426** comprise “fins” that extend upwards from the substrate surface **1428**. As the gate **1422** encompasses three sides of the semiconductor fin that extends from the source region **1424** to the drain region **1426**, the transistor **1420** can be considered a tri-gate transistor. FIG. **14**B illustrates one S/D fin extending through the gate **1422**, but multiple S/D fins can extend through the gate of a FinFET transistor.

(59) FIG. **14**C is a perspective view of a gate-all-around (GAA) transistor **1440** comprising a gate **1442** that controls current flow between a source region **1444** and a drain region **1446**. The transistor **1440** is non-planar in that the source region **1444** and the drain region **1446** are elevated

from the substrate surface **1428**.

(60) FIG. **14**D is a perspective view of a GAA transistor **1460** comprising a gate **1462** that controls current flow between multiple elevated source regions **1464** and multiple elevated drain regions **1466**. The transistor **1460** is a stacked GAA transistor as the gate controls the flow of current between multiple elevated S/D regions stacked on top of each other. The transistors **1440** and **1460** are considered gate-all-around transistors as the gates encompass all sides of the semiconductor portions that extends from the source regions to the drain regions. The transistors **1440** and **1460** can alternatively be referred to as nanowire, nanosheet, or nanoribbon transistors depending on the width (e.g., widths **1448** and **1468** of transistors **1440** and **1460**, respectively) of the semiconductor portions extending through the gate.

(61) Returning to FIG. **13**, a transistor **1340** may include a gate **1322** formed of at least two layers, a gate dielectric and a gate electrode. The gate dielectric may include one layer or a stack of layers. The one or more layers may include silicon oxide, silicon dioxide, silicon carbide, and/or a high-k dielectric material.

(62) The high-k dielectric material may include elements such as hafnium, silicon, oxygen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and zinc. Examples of high-k materials that may be used in the gate dielectric include, but are not limited to, hafnium oxide, hafnium silicon oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, and lead zinc niobate. In some embodiments, an annealing process may be carried out on the gate dielectric to improve its quality when a high-k material is used.

(63) The gate electrode may be formed on the gate dielectric and may include at least one p-type work function metal or n-type work function metal, depending on whether the transistor **1340** is to be a p-type metal oxide semiconductor (PMOS) or an n-type metal oxide semiconductor (NMOS) transistor. In some implementations, the gate electrode may consist of a stack of two or more metal layers, where one or more metal layers are work function metal layers and at least one metal layer is a fill metal layer. Further metal layers may be included for other purposes, such as a barrier layer.

(64) For a PMOS transistor, metals that may be used for the gate electrode include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, conductive metal oxides (e.g., ruthenium oxide), and any of the metals discussed below with reference to an NMOS transistor (e.g., for work function tuning). For an NMOS transistor, metals that may be used for the gate electrode include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, carbides of these metals (e.g., hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and aluminum carbide), and any of the metals discussed above with reference to a PMOS transistor (e.g., for work function tuning).

(65) In some embodiments, when viewed as a cross-section of the transistor **1340** along the source-channel-drain direction, the gate electrode may consist of a U-shaped structure that includes a bottom portion substantially parallel to the surface of the die substrate **1302** and two sidewall portions that are substantially perpendicular to the top surface of the die substrate **1302**. In other embodiments, at least one of the metal layers that form the gate electrode may simply be a planar layer that is substantially parallel to the top surface of the die substrate **1302** and does not include sidewall portions substantially perpendicular to the top surface of the die substrate **1302**. In other embodiments, the gate electrode may consist of a combination of U-shaped structures and planar, non-U-shaped structures. For example, the gate electrode may consist of one or more U-shaped metal layers formed atop one or more planar, non-U-shaped layers.

(66) In some embodiments, a pair of sidewall spacers may be formed on opposing sides of the gate stack to bracket the gate stack. The sidewall spacers may be formed from materials such as silicon nitride, silicon oxide, silicon carbide, silicon nitride doped with carbon, and silicon oxynitride. Processes for forming sidewall spacers are well known in the art and generally include deposition

and etching process steps. In some embodiments, a plurality of spacer pairs may be used; for instance, two pairs, three pairs, or four pairs of sidewall spacers may be formed on opposing sides of the gate stack.
(67) The S/D regions **1320** may be formed within the die substrate **1302** adjacent to the gate **1322** of individual transistors **1340**. The S/D regions **1320** may be formed using an implantation/diffusion process or an etching/deposition process, for example. In the former process, dopants such as boron, aluminum, antimony, phosphorous, or arsenic may be ion-implanted into the die substrate **1302** to form the S/D regions **1320**. An annealing process that activates the dopants and causes them to diffuse farther into the die substrate **1302** may follow the ion-implantation process. In the latter process, the die substrate **1302** may first be etched to form recesses at the locations of the S/D regions **1320**. An epitaxial deposition process may then be carried out to fill the recesses with material that is used to fabricate the S/D regions **1320**. In some implementations, the S/D regions **1320** may be fabricated using a silicon alloy such as silicon germanium or silicon carbide. In some embodiments, the epitaxially deposited silicon alloy may be doped in situ with dopants such as boron, arsenic, or phosphorous. In some embodiments, the S/D regions **1320** may be formed using one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. In further embodiments, one or more layers of metal and/or metal alloys may be used to form the S/D regions **1320**.
(68) Electrical signals, such as power and/or input/output (I/O) signals, may be routed to and/or from the devices (e.g., transistors **1340**) of the device layer **1304** through one or more interconnect layers disposed on the device layer **1304** (illustrated in FIG. **13** as interconnect layers **1306-1310**). For example, electrically conductive features of the device layer **1304** (e.g., the gate **1322** and the S/D contacts **1324**) may be electrically coupled with the interconnect structures **1328** of the interconnect layers **1306-1310**. The one or more interconnect layers **1306-1310** may form a metallization stack (also referred to as an “ILD stack”) **1319** of the integrated circuit device **1300**.
(69) The interconnect structures **1328** may be arranged within the interconnect layers **1306-1310** to route electrical signals according to a wide variety of designs; in particular, the arrangement is not limited to the particular configuration of interconnect structures **1328** depicted in FIG. **13**. Although a particular number of interconnect layers **1306-1310** is depicted in FIG. **13**, embodiments of the present disclosure include integrated circuit devices having more or fewer interconnect layers than depicted.
(70) In some embodiments, the interconnect structures **1328** may include lines **1328***a* and/or vias **1328***b* filled with an electrically conductive material such as a metal. The lines **1328***a* may be arranged to route electrical signals in a direction of a plane that is substantially parallel with a surface of the die substrate **1302** upon which the device layer **1304** is formed. For example, the lines **1328***a* may route electrical signals in a direction in and out of the page and/or in a direction across the page from the perspective of FIG. **13**. The vias **1328***b* may be arranged to route electrical signals in a direction of a plane that is substantially perpendicular to the surface of the die substrate **1302** upon which the device layer **1304** is formed. In some embodiments, the vias **1328***b* may electrically couple lines **1328***a* of different interconnect layers **1306-1310** together.
(71) The interconnect layers **1306-1310** may include a dielectric material **1326** disposed between the interconnect structures **1328**, as shown in FIG. **13**. In some embodiments, dielectric material **1326** disposed between the interconnect structures **1328** in different ones of the interconnect layers **1306-1310** may have different compositions; in other embodiments, the composition of the dielectric material **1326** between different interconnect layers **1306-1310** may be the same. The device layer **1304** may include a dielectric material **1326** disposed between the transistors **1340** and a bottom layer of the metallization stack as well. The dielectric material **1326** included in the device layer **1304** may have a different composition than the dielectric material **1326** included in the interconnect layers **1306-1310**; in other embodiments, the composition of the dielectric material **1326** in the device layer **1304** may be the same as a dielectric material **1326** included in any one of

the interconnect layers **1306-1310**.

(72) A first interconnect layer **1306** (referred to as Metal **1** or "M**1**") may be formed directly on the device layer **1304**. In some embodiments, the first interconnect layer **1306** may include lines **1328***a* and/or vias **1328***b*, as shown. The lines **1328***a* of the first interconnect layer **1306** may be coupled with contacts (e.g., the S/D contacts **1324**) of the device layer **1304**. The vias **1328***b* of the first interconnect layer **1306** may be coupled with the lines **1328***a* of a second interconnect layer **1308**.

(73) The second interconnect layer **1308** (referred to as Metal **2** or "M**2**") may be formed directly on the first interconnect layer **1306**. In some embodiments, the second interconnect layer **1308** may include via **1328***b* to couple the lines **1328** of the second interconnect layer **1308** with the lines **1328***a* of a third interconnect layer **1310**. Although the lines **1328***a* and the vias **1328***b* are structurally delineated with a line within individual interconnect layers for the sake of clarity, the lines **1328***a* and the vias **1328***b* may be structurally and/or materially contiguous (e.g., simultaneously filled during a dual-damascene process) in some embodiments.

(74) The third interconnect layer **1310** (referred to as Metal **3** or "M**3**") (and additional interconnect layers, as desired) may be formed in succession on the second interconnect layer **1308** according to similar techniques and configurations described in connection with the second interconnect layer **1308** or the first interconnect layer **1306**. In some embodiments, the interconnect layers that are "higher up" in the metallization stack **1319** in the integrated circuit device **1300** (i.e., farther away from the device layer **1304**) may be thicker that the interconnect layers that are lower in the metallization stack **1319**, with lines **1328***a* and vias **1328***b* in the higher interconnect layers being thicker than those in the lower interconnect layers.

(75) The integrated circuit device **1300** may include a solder resist material **1334** (e.g., polyimide or similar material) and one or more conductive contacts **1336** formed on the interconnect layers **1306-1310**. In FIG. **13**, the conductive contacts **1336** are illustrated as taking the form of bond pads. The conductive contacts **1336** may be electrically coupled with the interconnect structures **1328** and configured to route the electrical signals of the transistor(s) **1340** to external devices. For example, solder bonds may be formed on the one or more conductive contacts **1336** to mechanically and/or electrically couple an integrated circuit die including the integrated circuit device **1300** with another component (e.g., a printed circuit board). The integrated circuit device **1300** may include additional or alternate structures to route the electrical signals from the interconnect layers **1306-1310**; for example, the conductive contacts **1336** may include other analogous features (e.g., posts) that route the electrical signals to external components.

(76) In some embodiments in which the integrated circuit device **1300** is a double-sided die, the integrated circuit device **1300** may include another metallization stack (not shown) on the opposite side of the device layer(s) **1304**. This metallization stack may include multiple interconnect layers as discussed above with reference to the interconnect layers **1306-1310**, to provide conductive pathways (e.g., including conductive lines and vias) between the device layer(s) **1304** and additional conductive contacts (not shown) on the opposite side of the integrated circuit device **1300** from the conductive contacts **1336**.

(77) In other embodiments in which the integrated circuit device **1300** is a double-sided die, the integrated circuit device **1300** may include one or more through silicon vias (TSVs) through the die substrate **1302**; these TSVs may make contact with the device layer(s) **1304**, and may provide conductive pathways between the device layer(s) **1304** and additional conductive contacts (not shown) on the opposite side of the integrated circuit device **1300** from the conductive contacts **1336**. In some embodiments, TSVs extending through the substrate can be used for routing power and ground signals from conductive contacts on the opposite side of the integrated circuit device **1300** from the conductive contacts **1336** to the transistors **1340** and any other components integrated into the die **1300**, and the metallization stack **1319** can be used to route I/O signals from the conductive contacts **1336** to transistors **1340** and any other components integrated into the die **1300**.

(78) Multiple integrated circuit devices **1300** may be stacked with one or more TSVs in the individual stacked devices providing connection between one of the devices to any of the other devices in the stack. For example, one or more high-bandwidth memory (HBM) integrated circuit dies can be stacked on top of a base integrated circuit die and TSVs in the HBM dies can provide connection between the individual HBM and the base integrated circuit die. Conductive contacts can provide additional connections between adjacent integrated circuit dies in the stack. In some embodiments, the conductive contacts can be fine-pitch solder bumps (microbumps).
(79) FIG. **15** is a cross-sectional side view of an integrated circuit device assembly **1500** that may include any of the FeFET-based pbit devices disclosed herein. In some embodiments, the integrated circuit device assembly **1500** may be a microelectronic assembly. The integrated circuit device assembly **1500** includes a number of components disposed on a circuit board **1502** (which may be a motherboard, system board, mainboard, etc.). The integrated circuit device assembly **1500** includes components disposed on a first face **1540** of the circuit board **1502** and an opposing second face **1542** of the circuit board **1502**; generally, components may be disposed on one or both faces **1540** and **1542**.
(80) In some embodiments, the circuit board **1502** may be a printed circuit board (PCB) including multiple metal (or interconnect) layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. The individual metal layers comprise conductive traces. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **1502**. In other embodiments, the circuit board **1502** may be a non-PCB substrate. The integrated circuit device assembly **1500** illustrated in FIG. **15** includes a package-on-interposer structure **1536** coupled to the first face **1540** of the circuit board **1502** by coupling components **1516**. The coupling components **1516** may electrically and mechanically couple the package-on-interposer structure **1536** to the circuit board **1502**, and may include solder balls (as shown in FIG. **15**), pins (e.g., as part of a pin grid array (PGA), contacts (e.g., as part of a land grid array (LGA)), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.
(81) The package-on-interposer structure **1536** may include an integrated circuit component **1520** coupled to an interposer **1504** by coupling components **1518**. The coupling components **1518** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **1516**. Although a single integrated circuit component **1520** is shown in FIG. **15**, multiple integrated circuit components may be coupled to the interposer **1504**; indeed, additional interposers may be coupled to the interposer **1504**. The interposer **1504** may provide an intervening substrate used to bridge the circuit board **1502** and the integrated circuit component **1520**.
(82) The integrated circuit component **1520** may be a packaged or unpacked integrated circuit product that includes one or more integrated circuit dies (e.g., the die **1202** of FIG. **12**, the integrated circuit device **1300** of FIG. **13**) and/or one or more other suitable components. A packaged integrated circuit component comprises one or more integrated circuit dies mounted on a package substrate with the integrated circuit dies and package substrate encapsulated in a casing material, such as a metal, plastic, glass, or ceramic. In one example of an unpackaged integrated circuit component **1520**, a single monolithic integrated circuit die comprises solder bumps attached to contacts on the die. The solder bumps allow the die to be directly attached to the interposer **1504**. The integrated circuit component **1520** can comprise one or more computing system components, such as one or more processor units (e.g., system-on-a-chip (SoC), processor core, graphics processor unit (GPU), accelerator, chipset processor), I/O controller, memory, or network interface controller. In some embodiments, the integrated circuit component **1520** can comprise one or more additional active or passive devices such as capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electrostatic discharge (ESD) devices, and memory devices.

(83) In embodiments where the integrated circuit component **1520** comprises multiple integrated circuit dies, they dies can be of the same type (a homogeneous multi-die integrated circuit component) or of two or more different types (a heterogeneous multi-die integrated circuit component). A multi-die integrated circuit component can be referred to as a multi-chip package (MCP) or multi-chip module (MCM).
(84) In addition to comprising one or more processor units, the integrated circuit component **1520** can comprise additional components, such as embedded DRAM, stacked high bandwidth memory (HBM), shared cache memories, input/output (I/O) controllers, or memory controllers. Any of these additional components can be located on the same integrated circuit die as a processor unit, or on one or more integrated circuit dies separate from the integrated circuit dies comprising the processor units. These separate integrated circuit dies can be referred to as "chiplets". In embodiments where an integrated circuit component comprises multiple integrated circuit dies, interconnections between dies can be provided by the package substrate, one or more silicon interposers, one or more silicon bridges embedded in the package substrate (such as Intel® embedded multi-die interconnect bridges (EMIBs)), or combinations thereof.
(85) Generally, the interposer **1504** may spread connections to a wider pitch or reroute a connection to a different connection. For example, the interposer **1504** may couple the integrated circuit component **1520** to a set of ball grid array (BGA) conductive contacts of the coupling components **1516** for coupling to the circuit board **1502**. In the embodiment illustrated in FIG. **15**, the integrated circuit component **1520** and the circuit board **1502** are attached to opposing sides of the interposer **1504**; in other embodiments, the integrated circuit component **1520** and the circuit board **1502** may be attached to a same side of the interposer **1504**. In some embodiments, three or more components may be interconnected by way of the interposer **1504**.
(86) In some embodiments, the interposer **1504** may be formed as a PCB, including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. In some embodiments, the interposer **1504** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, an epoxy resin with inorganic fillers, a ceramic material, or a polymer material such as polyimide. In some embodiments, the interposer **1504** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The interposer **1504** may include metal interconnects **1508** and vias **1510**, including but not limited to through hole vias **1510-1** (that extend from a first face **1550** of the interposer **1504** to a second face **1554** of the interposer **1504**), blind vias **1510-2** (that extend from the first or second faces **1550** or **1554** of the interposer **1504** to an internal metal layer), and buried vias **1510-3** (that connect internal metal layers).
(87) In some embodiments, the interposer **1504** can comprise a silicon interposer. Through silicon vias (TSV) extending through the silicon interposer can connect connections on a first face of a silicon interposer to an opposing second face of the silicon interposer. In some embodiments, an interposer **1504** comprising a silicon interposer can further comprise one or more routing layers to route connections on a first face of the interposer **1504** to an opposing second face of the interposer **1504**.
(88) The interposer **1504** may further include embedded devices **1514**, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electrostatic discharge (ESD) devices, and memory devices. More complex devices such as radio frequency devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the interposer **1504**. The package-on-interposer structure **1536** may take the form of any of the package-on-interposer structures known in the art. In embodiments where the interposer is a non-printed circuit board
(89) The integrated circuit device assembly **1500** may include an integrated circuit component

1524 coupled to the first face **1540** of the circuit board **1502** by coupling components **1522**. The coupling components **1522** may take the form of any of the embodiments discussed above with reference to the coupling components **1516**, and the integrated circuit component **1524** may take the form of any of the embodiments discussed above with reference to the integrated circuit component **1520**.

(90) The integrated circuit device assembly **1500** illustrated in FIG. **15** includes a package-on-package structure **1534** coupled to the second face **1542** of the circuit board **1502** by coupling components **1528**. The package-on-package structure **1534** may include an integrated circuit component **1526** and an integrated circuit component **1532** coupled together by coupling components **1530** such that the integrated circuit component **1526** is disposed between the circuit board **1502** and the integrated circuit component **1532**. The coupling components **1528** and **1530** may take the form of any of the embodiments of the coupling components **1516** discussed above, and the integrated circuit components **1526** and **1532** may take the form of any of the embodiments of the integrated circuit component **1520** discussed above. The package-on-package structure **1534** may be configured in accordance with any of the package-on-package structures known in the art.

(91) FIG. **16** is a block diagram of an example electrical device **1600** that may include one or more of the FeFET-based pbit devices disclosed herein. For example, any suitable ones of the components of the electrical device **1600** may include one or more of the integrated circuit device assemblies **1500**, integrated circuit components **1520**, integrated circuit devices **1300**, or integrated circuit dies **1202** disclosed herein. A number of components are illustrated in FIG. **16** as included in the electrical device **1600**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some embodiments, some or all of the components included in the electrical device **1600** may be attached to one or more motherboards mainboards, or system boards. In some embodiments, one or more of these components are fabricated onto a single system-on-a-chip (SoC) die.

(92) Additionally, in various embodiments, the electrical device **1600** may not include one or more of the components illustrated in FIG. **16**, but the electrical device **1600** may include interface circuitry for coupling to the one or more components. For example, the electrical device **1600** may not include a display device **1606**, but may include display device interface circuitry (e.g., a connector and driver circuitry) to which a display device **1606** may be coupled. In another set of examples, the electrical device **1600** may not include an audio input device **1624** or an audio output device **1608**, but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **1624** or audio output device **1608** may be coupled.

(93) The electrical device **1600** may include one or more processor units **1602** (e.g., one or more processor units). As used herein, the terms "processor unit", "processing unit" or "processor" may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. The processor unit **1602** may include one or more digital signal processors (DSPs), application-specific integrated circuits (ASICs), central processing units (CPUs), graphics processing units (GPUs), general-purpose GPUs (GPGPUs), accelerated processing units (APUs), field-programmable gate arrays (FPGAs), neural network processing units (NPUs), data processor units (DPUs), accelerators (e.g., graphics accelerator, compression accelerator, artificial intelligence accelerator), controller cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, controllers, or any other suitable type of processor units. As such, the processor unit can be referred to as an XPU (or xPU).

(94) The electrical device **1600** may include a memory **1604**, which may itself include one or more memory devices such as volatile memory (e.g., dynamic random access memory (DRAM), static random-access memory (SRAM)), non-volatile memory (e.g., read-only memory (ROM), flash memory, chalcogenide-based phase-change non-voltage memories), solid state memory, and/or a

hard drive. In some embodiments, the memory **1604** may include memory that is located on the same integrated circuit die as the processor unit **1602**. This memory may be used as cache memory (e.g., Level 1 (L1), Level 2 (L2), Level 3 (L3), Level 4 (L4), Last Level Cache (LLC)) and may include embedded dynamic random access memory (eDRAM) or spin transfer torque magnetic random access memory (STT-MRAM).

(95) In some embodiments, the electrical device **1600** can comprise one or more processor units **1602** that are heterogeneous or asymmetric to another processor unit **1602** in the electrical device **1600**. There can be a variety of differences between the processing units **1602** in a system in terms of a spectrum of metrics of merit including architectural, microarchitectural, thermal, power consumption characteristics, and the like. These differences can effectively manifest themselves as asymmetry and heterogeneity among the processor units **1602** in the electrical device **1600**.

(96) In some embodiments, the electrical device **1600** may include a communication component **1612** (e.g., one or more communication components). For example, the communication component **1612** can manage wireless communications for the transfer of data to and from the electrical device **1600**. The term "wireless" and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term "wireless" does not imply that the associated devices do not contain any wires, although in some embodiments they might not.

(97) The communication component **1612** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultra mobile broadband (UMB) project (also referred to as "3GPP2"), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication component **1612** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication component **1612** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication component **1612** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication component **1612** may operate in accordance with other wireless protocols in other embodiments. The electrical device **1600** may include an antenna **1622** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

(98) In some embodiments, the communication component **1612** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., IEEE 802.3 Ethernet standards). As noted above, the communication component **1612** may include multiple communication components. For instance, a first communication component **1612** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication component **1612** may be dedicated to longer-range wireless communications such as global positioning system (GPS), EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some embodiments, a first communication component **1612** may be dedicated to wireless communications, and a second communication component **1612** may be dedicated to wired

communications.
(99) The electrical device **1600** may include battery/power circuitry **1614**. The battery/power circuitry **1614** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the electrical device **1600** to an energy source separate from the electrical device **1600** (e.g., AC line power).
(100) The electrical device **1600** may include a display device **1606** (or corresponding interface circuitry, as discussed above). The display device **1606** may include one or more embedded or wired or wirelessly connected external visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display.
(101) The electrical device **1600** may include an audio output device **1608** (or corresponding interface circuitry, as discussed above). The audio output device **1608** may include any embedded or wired or wirelessly connected external device that generates an audible indicator, such speakers, headsets, or earbuds.
(102) The electrical device **1600** may include an audio input device **1624** (or corresponding interface circuitry, as discussed above). The audio input device **1624** may include any embedded or wired or wirelessly connected device that generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output). The electrical device **1600** may include a Global Navigation Satellite System (GNSS) device **1618** (or corresponding interface circuitry, as discussed above), such as a Global Positioning System (GPS) device. The GNSS device **1618** may be in communication with a satellite-based system and may determine a geolocation of the electrical device **1600** based on information received from one or more GNSS satellites, as known in the art.
(103) The electrical device **1600** may include an other output device **1610** (or corresponding interface circuitry, as discussed above). Examples of the other output device **1610** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.
(104) The electrical device **1600** may include an other input device **1620** (or corresponding interface circuitry, as discussed above). Examples of the other input device **1620** may include an accelerometer, a gyroscope, a compass, an image capture device (e.g., monoscopic or stereoscopic camera), a trackball, a trackpad, a touchpad, a keyboard, a cursor control device such as a mouse, a stylus, a touchscreen, proximity sensor, microphone, a bar code reader, a Quick Response (QR) code reader, electrocardiogram (ECG) sensor, PPG (photoplethysmogram) sensor, galvanic skin response sensor, any other sensor, or a radio frequency identification (RFID) reader.
(105) The electrical device **1600** may have any desired form factor, such as a hand-held or mobile electrical device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a 2-in-1 convertible computer, a portable all-in-one computer, a netbook computer, an ultrabook computer, a personal digital assistant (PDA), an ultra mobile personal computer, a portable gaming console, etc.), a desktop electrical device, a server, a rack-level computing solution (e.g., blade, tray or sled computing systems), a workstation or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a stationary gaming console, smart television, a vehicle control unit, a digital camera, a digital video recorder, a wearable electrical device or an embedded computing system (e.g., computing systems that are part of a vehicle, smart home appliance, consumer electronics product or equipment, manufacturing equipment). In some embodiments, the electrical device **1600** may be any other electronic device that processes data. In some embodiments, the electrical device **1600** may comprise multiple discrete physical components. Given the range of devices that the electrical device **1600** can be manifested as in various embodiments, in some embodiments, the electrical device **1600** can be referred to as a computing device or a computing system.
(106) Some examples of embodiments are provided below. As used in the following examples, the

term “connected” may refer to an electrical connection. In some instances, the connection may be a direct connection between two items/components. Further, as used in the following examples, the term “coupled” may refer to a connection that may be direct or indirect. For example, a first component coupled to a second component may include a third component connected between the first and second components.
(107) Example 1 includes an apparatus comprising: a first field-effect transistor (FET) comprising: a source region; a drain region; a source electrode on the source region; a drain electrode on the drain region; a channel region between the source and drain regions; a dielectric layer on a surface over the channel region; an electrode layer above the dielectric layer; and a ferroelectric (FE) material layer between the dielectric layer and the electrode layer; and a second FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the second FET is connected to the drain electrode of the first FET.
(108) Example 2 includes the subject matter of Example 1, further comprising: a third FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the third FET is connected to a first supply voltage terminal, the gate electrode of the third FET is connected to a first clock signal terminal, and the drain electrode of the third FET is connected to the electrode layer of the first FET; a fourth FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the fourth FET is connected to a second supply voltage terminal, the gate electrode of the fourth FET is connected to a second clock signal terminal, and the drain electrode of the fourth FET is connected to the electrode layer of the first FET; and a fifth FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the gate electrode of the fifth FET is connected to a third clock signal terminal, and the drain electrode of the fifth FET is connected to the electrode layer of the first FET.
(109) Example 3 includes the subject matter of Example 1, wherein the electrode layer is a first electrode layer, and the first FET further comprises a second electrode layer between the FE material layer and the dielectric layer.
(110) Example 4 includes the subject matter of Example 3, further comprising: a third FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the third FET is connected to a first supply voltage terminal, the gate electrode of the third FET is connected to a first clock signal terminal, and the drain electrode of the third FET is connected to the electrode layer of the first FET; a fourth FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the fourth FET is connected to a second supply voltage terminal, the gate electrode of the fourth FET is connected to a second clock signal terminal, and the drain electrode of the fourth FET is connected to the electrode layer of the first FET; and a fifth FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the fifth FET is connected to a third supply voltage terminal, the gate electrode of the fifth FET is connected to a third clock signal terminal, and the drain electrode of the fifth FET is connected to the electrode layer of the first FET.
(111) Example 5 includes the subject matter of Example 3 or 4, further comprising a sixth FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the gate electrode of the sixth FET is connected to the third clock signal terminal, and the drain electrode of the third FET is connected to the second electrode layer of the first FET.
(112) Example 6 includes the subject matter of any one of Examples 1-5, further comprising a D flip-flip circuit connected to the drain electrodes of the first and second FETs.
(113) Example 7 includes the subject matter of any one of Examples 1-6, wherein the first FET is an n-channel FET and the source electrode of the first FET is connected to a ground terminal, the second FET is a p-channel FET and the source electrode of the second FET is connected to a supply voltage terminal.
(114) Example 8 includes the subject matter of any one of Examples 1-6, wherein the first FET is a p-channel FET and the source electrode of the first FET is connected to a supply voltage terminal,

and the second FET is an n-channel FET and the source electrode of the second FET is connected to a ground terminal.
(115) Example 9 includes the subject matter of any one of Examples 1-8, wherein the FE material comprises hafnium and oxygen.
(116) Example 10 includes the subject matter of any one of Examples 1-9, wherein the first FET is a planar FET.
(117) Example 11 includes the subject matter of any one of Examples 1-9, wherein the first FET is a non-planar FET.
(118) Example 12 includes a system comprising: a plurality of pbit devices, each pbit device comprising: a field-effect transistor (FET) comprising: a source region; a drain region; a source electrode on the source region; a drain electrode on the drain region; a channel region between the source and drain regions; and a layer stack on a surface of the FET over the channel region, the layer stack comprising a dielectric layer, an electrode layer, and a ferroelectric (FE) material layer; an input node coupled to the electrode layer of the FET; and an output node coupled to the drain electrode of the FET; wherein the input node of each pbit device is coupled to an output node of one or more other pbit devices.
(119) Example 13 includes the subject matter of Example 12, wherein the input node of a first pbit device is coupled to output nodes of a second pbit device and a third pbit device through a circuit comprising: a first resistor connected to the output node of the second pbit device; a second resistor connected to the output node of the third pbit device; and an opamp circuit comprising a positive input terminal, a negative input terminal, and an output terminal; wherein the positive input terminal of the opamp circuit is connected to a reference voltage source, the negative input terminal the opamp circuit is connected to the first and second resistors and to the output terminal via a third resistor, and the output terminal is coupled to the input node of the first pbit device.
(120) Example 14 includes the subject matter of Example 13, wherein the first and second resistor are configurable resistors.
(121) Example 15 includes the subject matter of Example 12, wherein the input node of a first pbit device is coupled to output nodes of a second pbit device and a third pbit device through a circuit comprising: a first capacitor connected to the output node of the second pbit device; a second capacitor connected to the output node of the third pbit device; and an opamp circuit comprising a positive input terminal, a negative input terminal, and an output terminal; wherein the positive input terminal of the opamp circuit is connected to a reference voltage source, the negative input terminal the opamp circuit is connected to the first and second capacitors and to the output terminal via a third capacitor and a FET is parallel, and the output terminal is coupled to the input node of the first pbit device.
(122) Example 16 includes the subject matter of any one of Examples 12-15, wherein each pbit device further comprises a D flip-flop circuit, the drain electrode of the FET is connected to an input of the D flip-flop circuit, and the output node of the pbit device is connected to an output of the D flip-flop circuit.
(123) Example 17 includes the subject matter of any one of Examples 12-16, wherein the layer stack FET of each pbit device comprises the FE material layer between the dielectric layer and the electrode layer.
(124) Example 18 includes the subject matter of any one of Examples 12-17, further comprising pulse generation circuitry comprising: a first voltage source; a second voltage source; a first clock signal generator; a second clock signal generator; a third clock signal generator; a first FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the first FET is connected to the first voltage source, the gate electrode of the first FET is connected to the first clock signal generator, and the drain electrode of the first FET is connected to the input node of each pbit device; a second FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the second FET is connected to the second

voltage source, the gate electrode of the first FET is connected to the second clock signal generator, and the drain electrode of the first FET is connected to the input node of each pbit device; and a third FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the third FET is coupled to the output node of one or more other pbit devices, the gate electrode of the third FET is connected to the third clock signal generator, and the drain electrode of the first FET is connected to the input node of each pbit device.
(125) Example 19 includes the subject matter of any one of Examples 12-16, wherein the electrode layer of at least one pbit device is a first electrode layer, the layer stack of the at least one pbit device further comprises a second electrode layer, the FE material is between the first and second electrode layers.
(126) Example 20 includes the subject matter of Example 19, further comprising pulse generation circuitry comprising: a first voltage source; a second voltage source; a third voltage source; a first clock signal generator; a second clock signal generator; a third clock signal generator; a first FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the first FET is connected to the first voltage source, the gate electrode of the first FET is connected to the first clock signal generator, and the drain electrode of the first FET is connected to the second electrode layer of each pbit device; a second FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the second FET is connected to the second voltage source, the gate electrode of the first FET is connected to the second clock signal generator, and the drain electrode of the first FET is connected to the second electrode layer of each pbit device; and a third FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the third FET is coupled to the third voltage source, the gate electrode of the third FET is connected to the third clock signal generator, and the drain electrode of the first FET is connected to the second electrode layer of each pbit device.
(127) Example 21 includes the subject matter of any one of Examples 12-20, wherein the FET of each pbit device is a first FET of the pbit device, and each pbit device further comprises a second FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the second FET is connected to the drain electrode of the first FET.
(128) Example 22 includes a method of operating a field-effect transistor (FET) with a ferroelectric (FE) material layer, the method comprising: applying a number of pulse cycles to a gate electrode of the FET, each pulse cycle comprising: a reset pulse during a first time period, wherein a magnitude of the reset pulse is to deterministically set a polarization state of the FE material layer of the FET; a write pulse during a second time period after the first time period; and a read pulse during a third time period after the third time period, wherein a magnitude of the read pulse is to be between a first threshold voltage of the FET based on a first polarization state of the FE material layer and a second threshold voltage of the FET based on a second polarization state of the FE material layer; and determining, for each pulse cycle, a polarization state of the FE material layer based on the read pulse of the pulse cycle.
(129) Example 23 includes the subject matter of Example 22, further comprising determining a statistical probability of the polarization state of the FE material layer after the number of pulse cycles.
(130) Example 24 includes the subject matter of Example 22, wherein a magnitude of each write pulse is based on an output voltage of at least one other FET with a FE material layer, the output voltage of the at least one other FET based on a polarization state of the FE material layer of the FET.
(131) Example 25 includes the subject matter of Example 24, wherein the magnitude of each write pulse is based on a sum of the output voltages.
(132) Example 26 includes a computer-readable medium comprising instructions that, when executed by a machine, cause the machine to implement the method of any one of Examples 22-25.
(133) Example 27 includes an apparatus comprising means to implement the method of any one of

Examples 22-25.
(134) In the foregoing, a detailed description has been given with reference to specific example embodiments. It will, however, be evident that various modifications and changes may be made thereto without departing from the broader spirit and scope of the disclosure as set forth in the appended claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense. Furthermore, the foregoing use of embodiment(s) and other exemplarily language does not necessarily refer to the same embodiment or the same example, but may refer to different and distinct embodiments, as well as potentially the same embodiment.

Claims

1. An apparatus comprising: a first field-effect transistor (FET) comprising: a source region; a drain region; a source electrode on the source region; a drain electrode on the drain region; a channel region between the source and drain regions; a dielectric layer on a surface over the channel region; an electrode layer above the dielectric layer; and a ferroelectric (FE) material layer between the dielectric layer and the electrode layer; a second FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the second FET is connected to the drain electrode of the first FET; and a D flip-flop circuit connected to the drain electrode of the first FET and the drain electrode of the second FET.
2. The apparatus of claim 1, further comprising: a third FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the third FET is connected to a first supply voltage terminal, the gate electrode of the third FET is connected to a first clock signal terminal, and the drain electrode of the third FET is connected to the electrode layer of the first FET; a fourth FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the fourth FET is connected to a second supply voltage terminal, the gate electrode of the fourth FET is connected to a second clock signal terminal, and the drain electrode of the fourth FET is connected to the electrode layer of the first FET; and a fifth FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the gate electrode of the fifth FET is connected to a third clock signal terminal, and the drain electrode of the fifth FET is connected to the electrode layer of the first FET.
3. The apparatus of claim 1, wherein the electrode layer is a first electrode layer, and the first FET further comprises a second electrode layer between the FE material layer and the dielectric layer.
4. The apparatus of claim 3, further comprising a third FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the third FET is connected to a first supply voltage terminal, the gate electrode of the third FET is connected to a first clock signal terminal, and the drain electrode of the third FET is connected to the electrode layer of the first FET; a fourth FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the fourth FET is connected to a second supply voltage terminal, the gate electrode of the fourth FET is connected to a second clock signal terminal, and the drain electrode of the fourth FET is connected to the electrode layer of the first FET; and a fifth FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the fifth FET is connected to a third supply voltage terminal, the gate electrode of the fifth FET is connected to a third clock signal terminal, and the drain electrode of the fifth FET is connected to the electrode layer of the first FET.
5. The apparatus of claim 4, further comprising a sixth FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the gate electrode of the sixth FET is connected to the third clock signal terminal, and the drain electrode of the third FET is connected to the second electrode layer of the first FET.
6. The apparatus of claim 1, wherein the first FET is an n-channel FET and the source electrode of the first FET is connected to a ground terminal, the second FET is a p-channel FET and the source

electrode of the second FET is connected to a supply voltage terminal.
7. The apparatus of claim 1, wherein the first FET is a p-channel FET and the source electrode of the first FET is connected to a supply voltage terminal, and the second FET is an n-channel FET and the source electrode of the second FET is connected to a ground terminal.
8. The apparatus of claim 1, wherein the FE material comprises hafnium and oxygen.
9. The apparatus of claim 1, wherein the first FET is a planar FET.
10. The apparatus of claim 1, wherein the first FET is a non-planar FET.
11. A system comprising: a plurality of pbit devices, each pbit device comprising: a field-effect transistor (FET) comprising: a source region; a drain region; a source electrode on the source region; a drain electrode on the drain region; a channel region between the source and drain regions; and a layer stack on a surface of the FET over the channel region, the layer stack comprising a dielectric layer, an electrode layer, and a ferroelectric (FE) material layer; an input node coupled to the electrode layer of the FET; and an output node coupled to the drain electrode of the FET; wherein the input node of each pbit device is coupled to an output node of one or more other pbit devices.
12. The system of claim 11, wherein the input node of a first pbit device is coupled to output nodes of a second pbit device and a third pbit device through a circuit comprising: a first resistor connected to the output node of the second pbit device; a second resistor connected to the output node of the third pbit device; and an opamp circuit comprising a positive input terminal, a negative input terminal, and an output terminal; wherein the positive input terminal of the opamp circuit is connected to a reference voltage source, the negative input terminal the opamp circuit is connected to the first and second resistors and to the output terminal via a third resistor, and the output terminal is coupled to the input node of the first pbit device.
13. The system of claim 12, wherein the first and second resistor are configurable resistors.
14. The system of claim 11, wherein the input node of a first pbit device is coupled to output nodes of a second pbit device and a third pbit device through a circuit comprising: a first capacitor connected to the output node of the second pbit device; a second capacitor connected to the output node of the third pbit device; and an opamp circuit comprising a positive input terminal, a negative input terminal, and an output terminal; wherein the positive input terminal of the opamp circuit is connected to a reference voltage source, the negative input terminal the opamp circuit is connected to the first and second capacitors and to the output terminal via a third capacitor and a FET is parallel, and the output terminal is coupled to the input node of the first pbit device.
15. The system of claim 11, wherein each pbit device further comprises a D flip-flop circuit, the drain electrode of the FET is connected to an input of the D flip-flop circuit, and the output node of the pbit device is connected to an output of the D flip-flop circuit.
16. The system of claim 11, wherein the layer stack FET of each pbit device comprises the FE material layer between the dielectric layer and the electrode layer.
17. The system of claim 11, further comprising pulse generation circuitry comprising: a first voltage source; a second voltage source; a first clock signal generator; a second clock signal generator; a third clock signal generator; a first FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the first FET is connected to the first voltage source, the gate electrode of the first FET is connected to the first clock signal generator, and the drain electrode of the first FET is connected to the input node of each pbit device; a second FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the second FET is connected to the second voltage source, the gate electrode of the first FET is connected to the second clock signal generator, and the drain electrode of the first FET is connected to the input node of each pbit device; and a third FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the third FET is coupled to the output node of one or more other pbit devices, the gate electrode of the third FET is connected to the third clock signal generator, and the drain electrode of the first FET is connected to the input

node of each pbit device.
18. The system of claim 11, wherein the electrode layer of at least one pbit device is a first electrode layer, the layer stack of the at least one pbit device further comprises a second electrode layer, the FE material is between the first and second electrode layers.
19. The system of claim 18, further comprising pulse generation circuitry comprising: a first voltage source; a second voltage source; a third voltage source; a first clock signal generator; a second clock signal generator; a third clock signal generator; a first FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the first FET is connected to the first voltage source, the gate electrode of the first FET is connected to the first clock signal generator, and the drain electrode of the first FET is connected to the second electrode layer of each pbit device; a second FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the second FET is connected to the second voltage source, the gate electrode of the first FET is connected to the second clock signal generator, and the drain electrode of the first FET is connected to the second electrode layer of each pbit device; and a third FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the third FET is coupled to the third voltage source, the gate electrode of the third FET is connected to the third clock signal generator, and the drain electrode of the first FET is connected to the second electrode layer of each pbit device.
20. The system of claim 11, wherein the FET of each pbit device is a first FET of the pbit device, and each pbit device further comprises a second FET comprising a source electrode, a drain electrode, and a gate electrode, wherein the drain electrode of the second FET is connected to the drain electrode of the first FET.
21. A method of operating a field-effect transistor (FET) with a ferroelectric (FE) material layer, the method comprising: applying a number of pulse cycles to a gate electrode of the FET, each pulse cycle comprising: a reset pulse during a first time period, wherein a magnitude of the reset pulse is to deterministically set a polarization state of the FE material layer of the FET; a write pulse during a second time period after the first time period; and a read pulse during a third time period after the third time period, wherein a magnitude of the read pulse is to be between a first threshold voltage of the FET based on a first polarization state of the FE material layer and a second threshold voltage of the FET based on a second polarization state of the FE material layer; and determining, for each pulse cycle, a polarization state of the FE material layer based on the read pulse of the pulse cycle.
22. The method of claim 21, further comprising determining a statistical probability of the polarization state of the FE material layer after the number of pulse cycles.
23. The method of claim 21, wherein a magnitude of each write pulse is based on an output voltage of at least one other FET with a FE material layer, the output voltage of the at least one other FET based on a polarization state of the FE material layer of the FET.
24. The method of claim 23, wherein the magnitude of each write pulse is based on a sum of the output voltages.